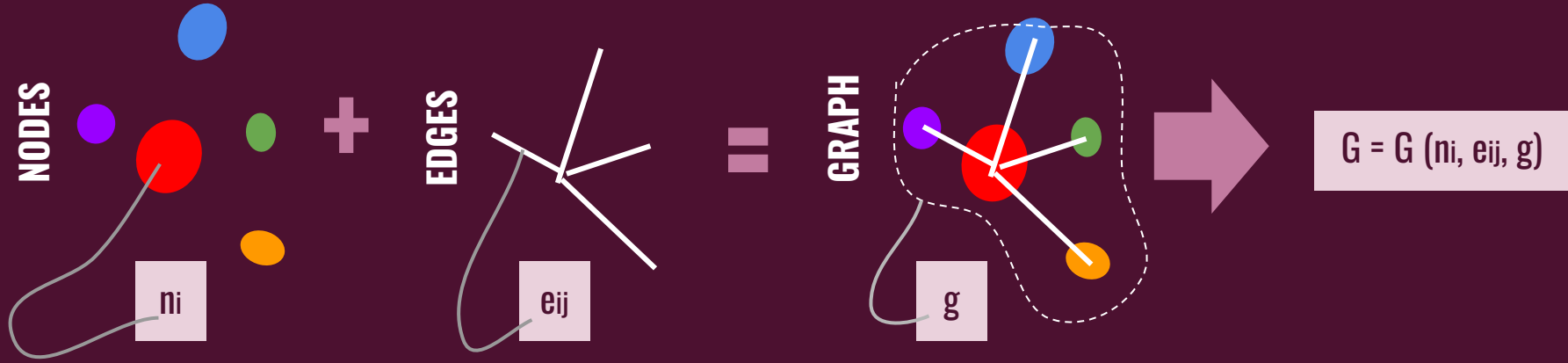


HOW TO USE GRAPH NEURAL NETWORKS TO UNDERSTAND THE UNIVERSE?

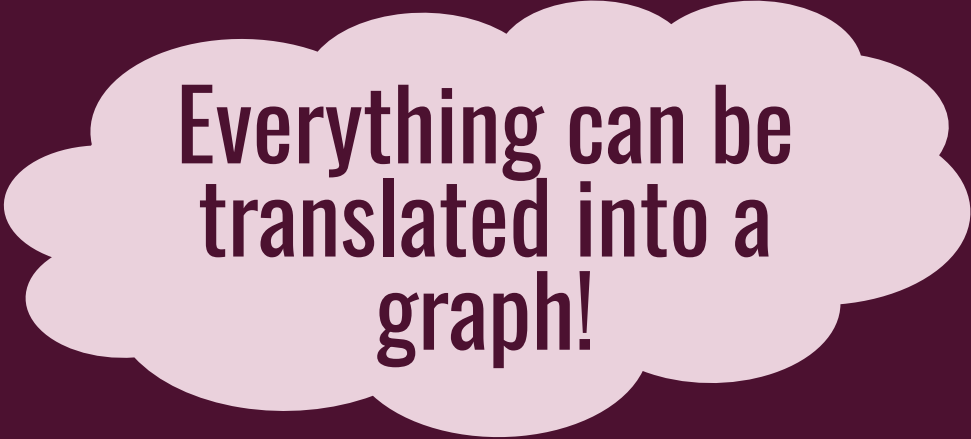
Ph.D. Natalí S. M. de Santi

AstroGainz
June 19th, 2025

What are graphs?



What kind of data can be seen as graphs?



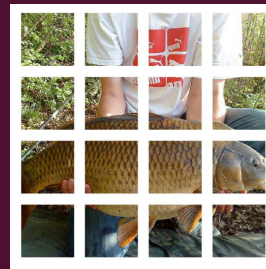
**Everything can be
translated into a
graph!**

What kind of data can be seen as graphs?

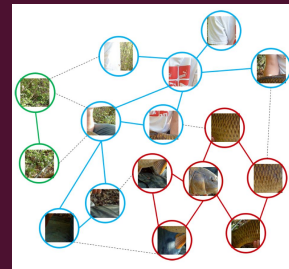
Tabular data

X	...	Y
15.7	...	24.2
12.1	...	14.6
...	...	...
13.0	...	15.9

Images



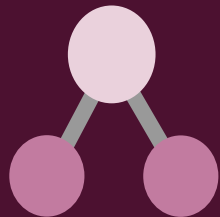
Grid structure



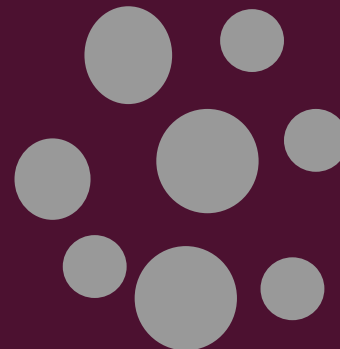
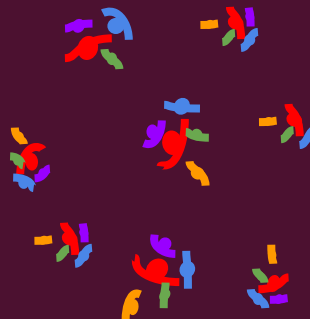
Graph structure

[arxiv:2206.00272](https://arxiv.org/abs/2206.00272)

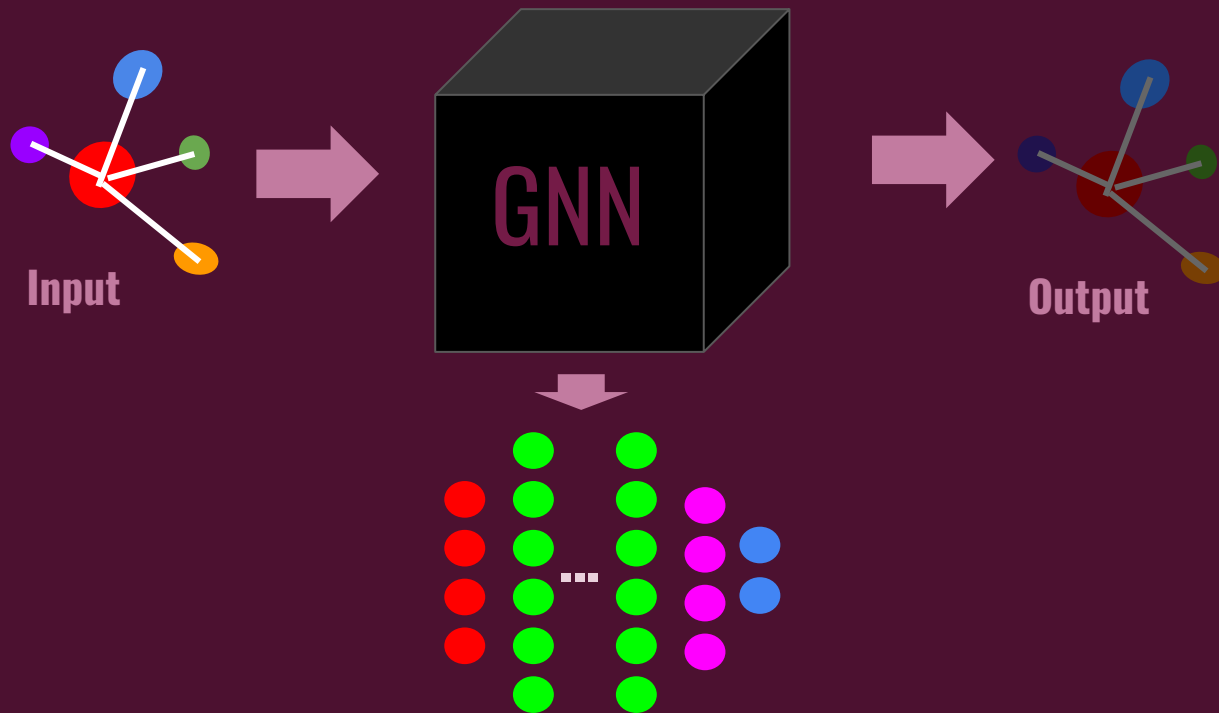
Molecules



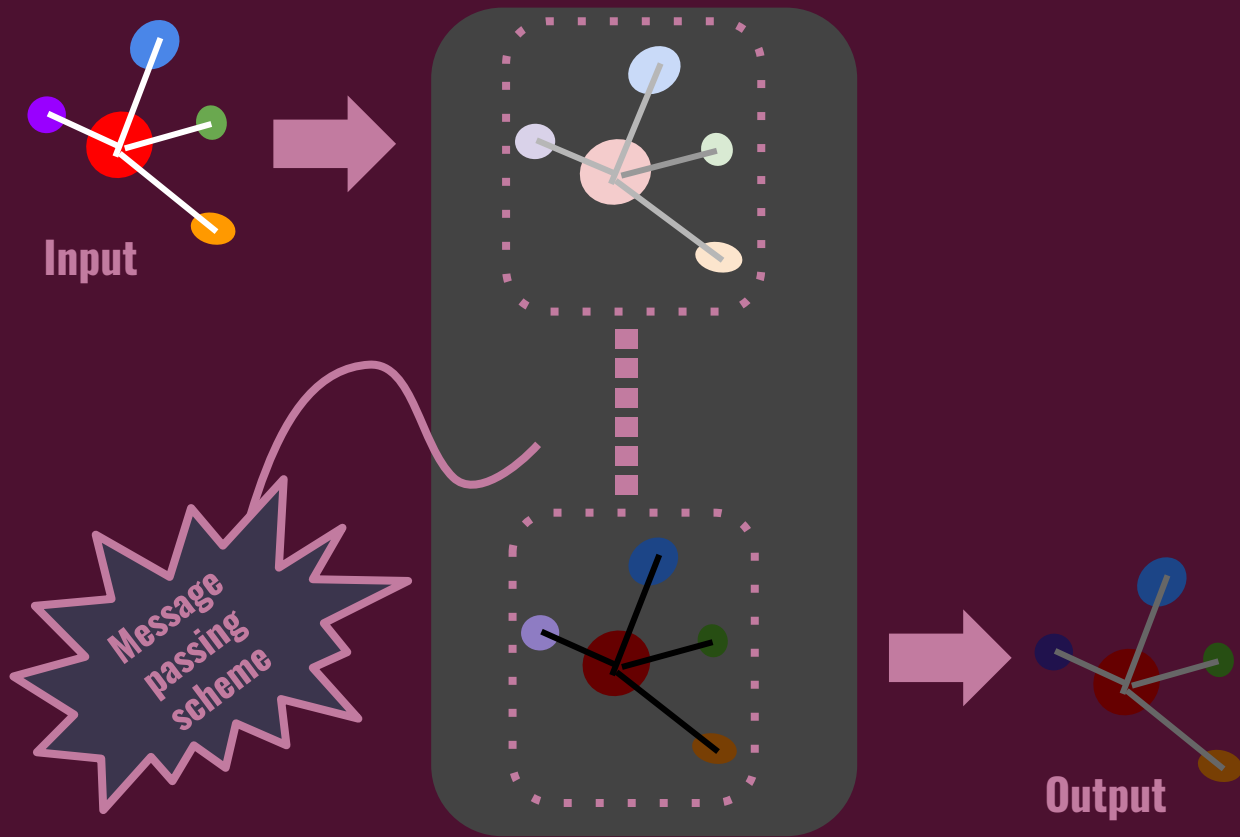
Galaxies & Halos



What are Graph Neural Networks?

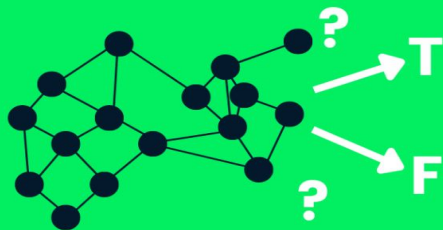


What is the basic principle behind Graph Neural Networks?

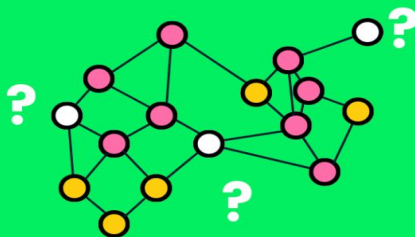


Are GNNs used for doing what?

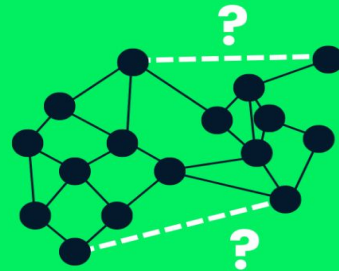
Graph Classification



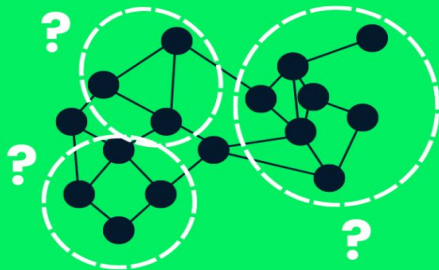
Node Classification



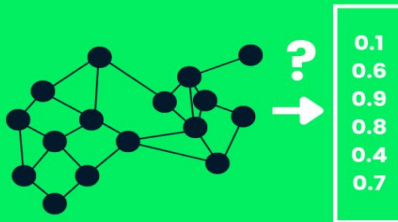
Link Prediction



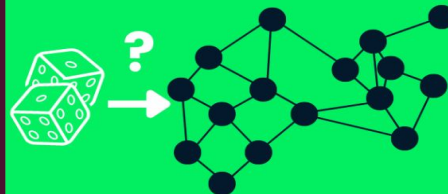
Community Detection



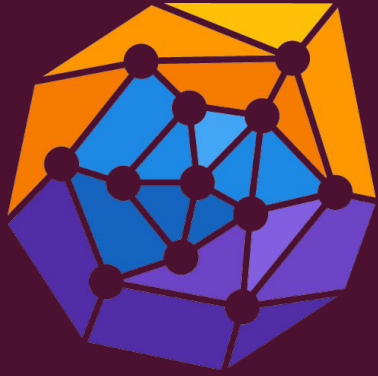
Graph Embedding



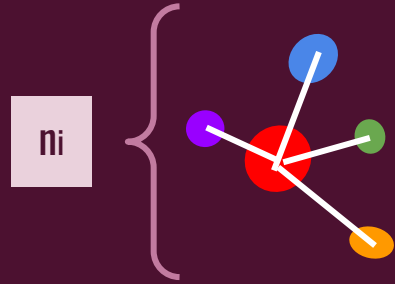
Graph Generation



Different Graph Neural Network Architectures

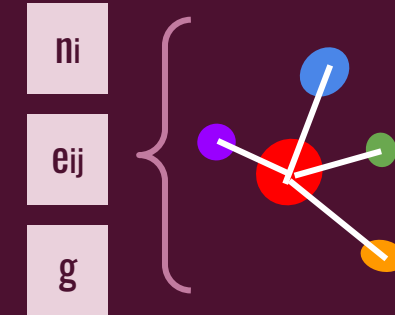


PyTorch Geometric



GRAPH CONVOLUTIONAL NETWORKS

GRAPH SAGE NETWORKS



GRAPH NEURAL NETWORKS FOR COSMOLOGY

Different Graph Neural Network Architectures

GRAPH CONVOLUTIONAL NETWORKS

Designed to perform a convolution operation (matrix multiplication) on graphs using node attributes

Graph Convolutional Network Layer

$$\mathbf{n}_i^{\ell+1} = \sum_{j \in \mathfrak{N}_{(i)} \cup i} \frac{1}{\sqrt{\deg(i)} \cdot \sqrt{\deg(j)}} \cdot \left[\mathbf{W}^T \cdot \mathbf{n}_j^{(\ell)} \right] + \mathbf{b}$$

Learnable
weight matrix

Additive
bias

Edge Convolutional Layer

$$\mathbf{n}_i^{(\ell+1)} = \max_{j \in \mathfrak{N}_j} \mathcal{H}^{(\ell+1)} \left([\mathbf{n}_i^\ell, \mathbf{n}_j^\ell - \mathbf{n}_i^\ell] \right)$$

Aggregation
Function

MLP

"Edge"
information

Different Graph Neural Network Architectures

GRAPH SAGE NETWORKS

Designed to update node information on graphs based on the neighborhood of each node

SAGE Convolutional Layer

$$\mathbf{n}_i^{\ell+1} = \mathbf{W}_1 \mathbf{n}_i^{\ell} + \mathbf{W}_2 \cdot \text{mean}_{j \in \mathfrak{N}_i} \mathbf{n}_j^{\ell}$$

Learnable
weight matrices

Different Graph Neural Network Architectures

GRAPH NEURAL NETWORKS FOR COSMOLOGY

Meta Layer

Edge model

$$\mathbf{e}_{ij}^{(\ell+1)} = \mathcal{E}^{(\ell+1)} \left(\left[\mathbf{n}_i^{(\ell)}, \mathbf{n}_j^{(\ell)}, \mathbf{e}_{ij}^{(\ell)} \right] \right)$$

MLP

Node model

$$\mathbf{n}_i^{(\ell+1)} = \mathcal{N}^{(\ell+1)} \left(\left[\mathbf{n}_i^{(\ell)}, \bigoplus_{j \in \mathfrak{N}_i} \mathbf{e}_{ij}^{(\ell+1)}, \mathbf{g} \right] \right)$$

MLP

Multi Pooling Operation

$$\bigoplus_{j \in \mathfrak{N}_i} \mathbf{e}_{ij}^{(\ell+1)} = \left[\max_{j \in \mathfrak{N}_i} \mathbf{e}_{ij}^{(\ell+1)}, \sum_{j \in \mathfrak{N}_i} \mathbf{e}_{ij}^{(\ell+1)}, \frac{\sum_{j \in \mathfrak{N}_i} \mathbf{e}_{ij}^{(\ell+1)}}{\sum_{j \in \mathfrak{N}_i} 1} \right]$$

Different Graph Neural Network Architectures

GRAPH NEURAL NETWORKS FOR COSMOLOGY

Deep Sets

- Operate on sets of **unordered data**
- Similarly to MLPs, but take into account **permutation invariance**

$$\mathbf{n}_i^{(\ell+1)} = \mathcal{N}^{(\ell+1)} \left(\mathbf{n}_i^{(\ell)} \right)$$

No Initial Node Attributes

Edge Model:

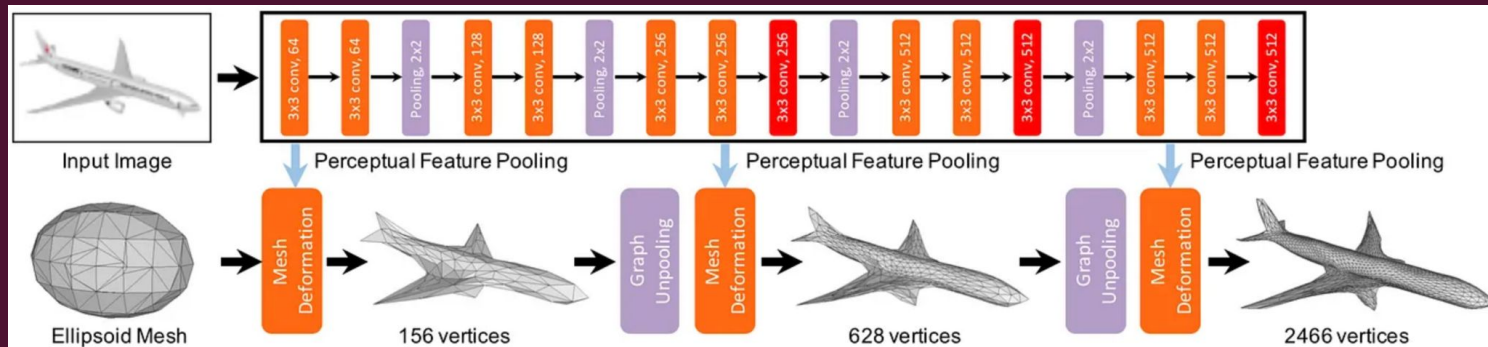
$$\mathbf{e}_{ij}^{(1)} = \mathcal{E}^{(1)} \left(\mathbf{e}_{ij}^{(0)} \right)$$

Node Model:

$$\mathbf{n}_i^{(1)} = \mathcal{N}^{(1)} \left(\left[\bigoplus_{j \in \mathfrak{N}_i} \mathbf{e}_{ij}^{(1)}, \mathbf{g}^{(0)} \right] \right)$$

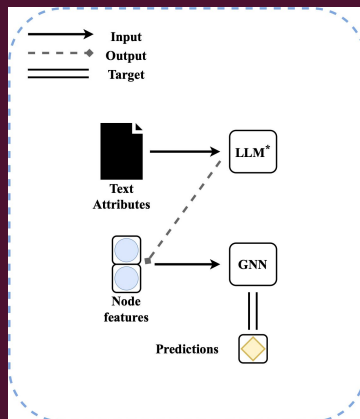
Different Graph Neural Network Applications

Computer Vision



Pixel2Mesh: Generating 3D Mesh Models from Single RGB Images [[arXiv: 1804.01654](https://arxiv.org/abs/1804.01654)]

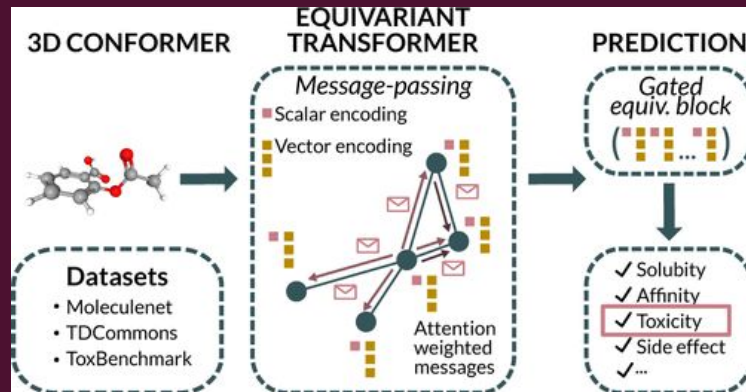
Large Language Models



Exploring the Potential of Large Language Models (LLMs) in Learning on Graphs [[arXiv: 2307.03393](https://arxiv.org/abs/2307.03393)]

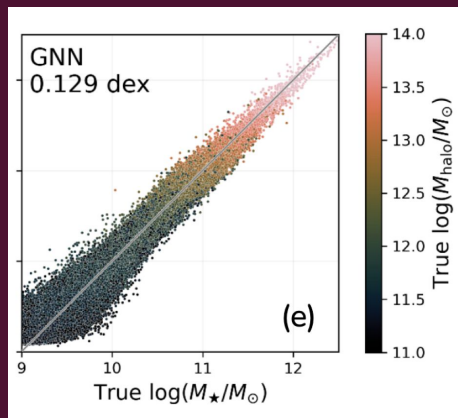
Different Graph Neural Network Applications

Chemistry

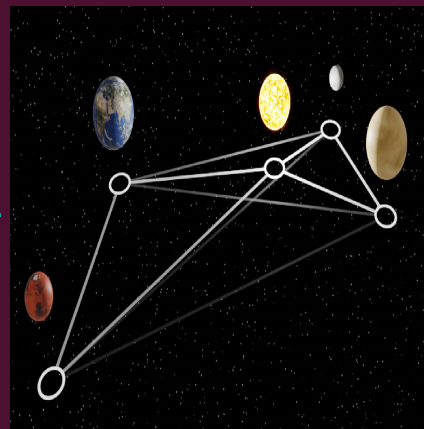


Equivariant Graph Neural Networks for Toxicity Prediction
[Cremer+2023]

Astrophysics



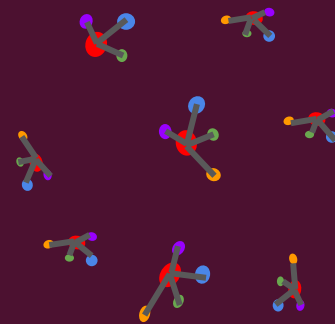
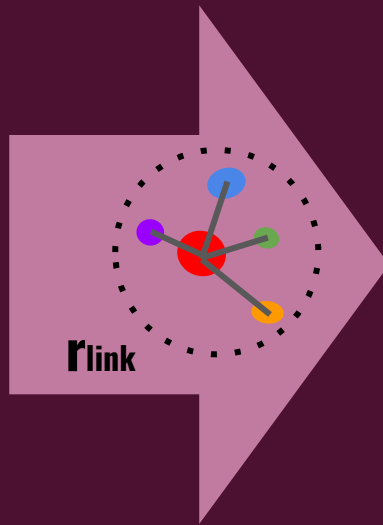
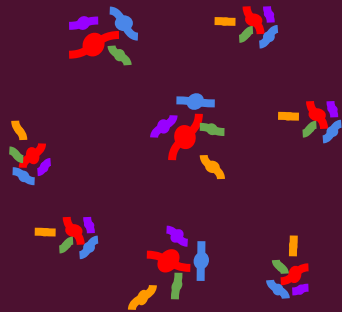
Halo-galaxy connection [arXiv: 2306.12327]



Rediscovering orbital mechanics with machine learning
[arXiv: 2202.02306]

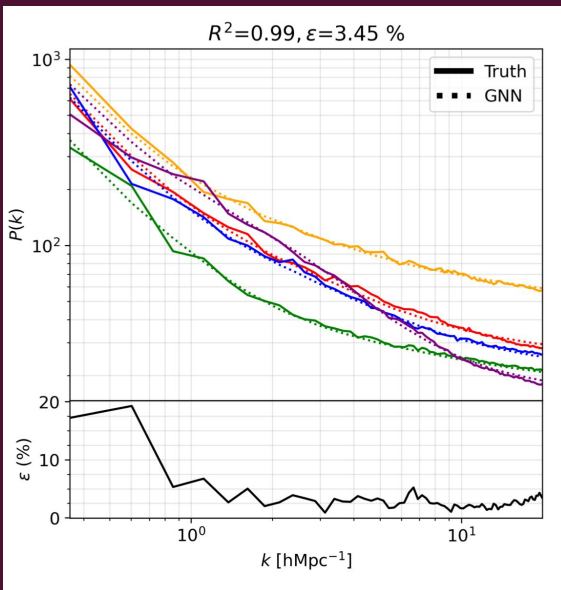
Different Graph Neural Network Applications

Cosmology

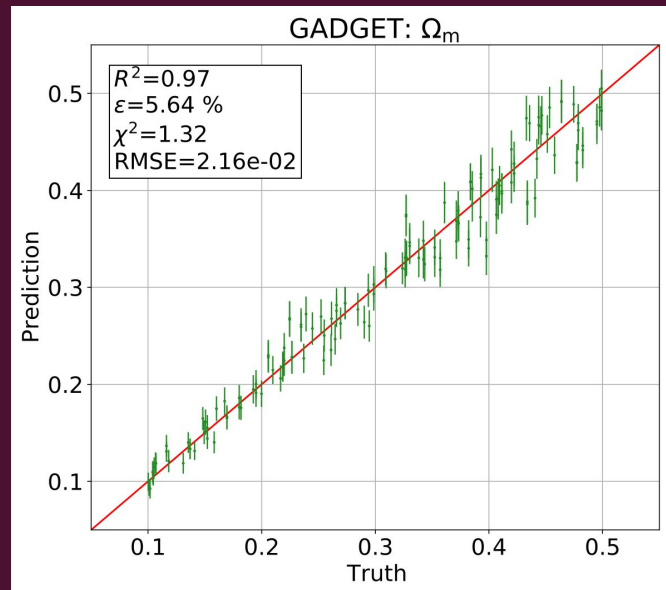


Natural representation of the
cosmic-web

Cosmology

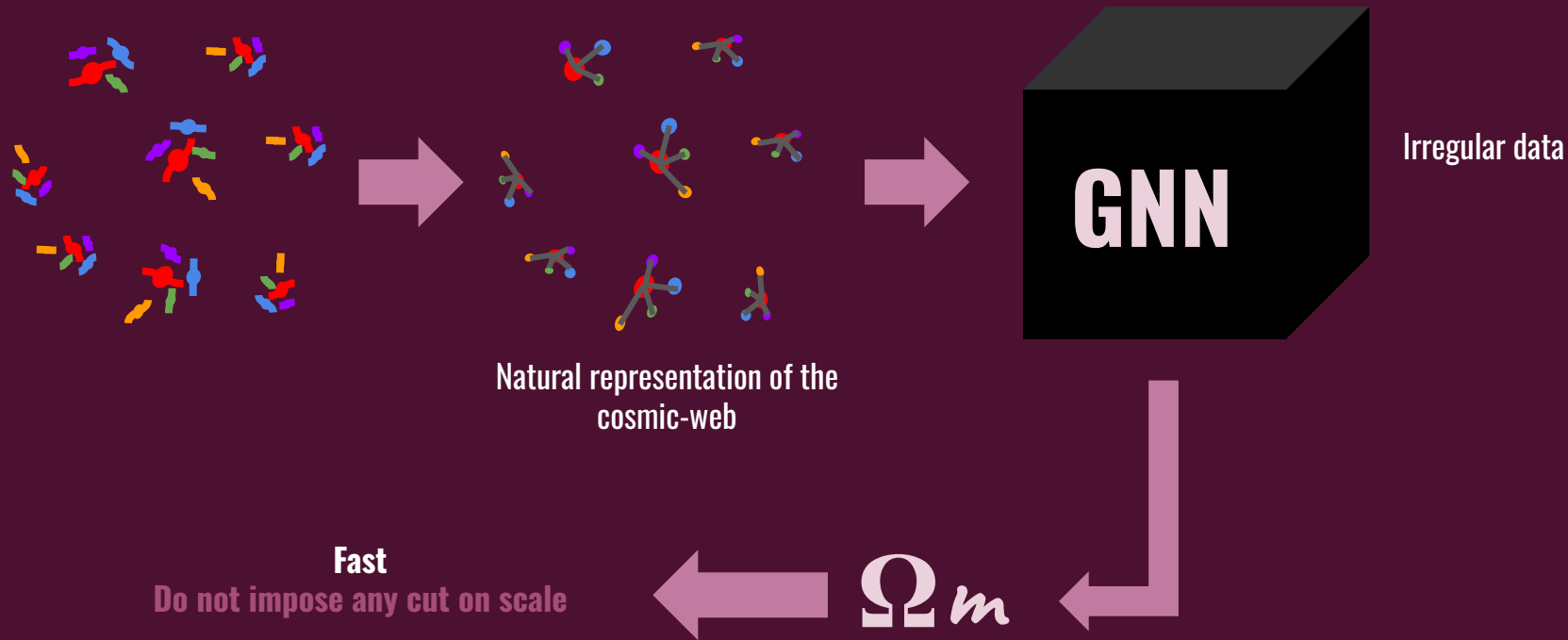


GNNs to compute the power spectrum from galaxies [[arXiv: 2204.13713](#)]



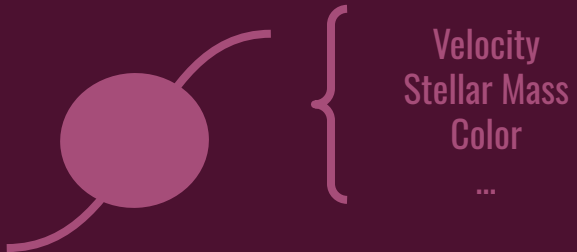
GNNs to predict the matter content of the universe using halos [[arXiv: 2209.06843](#)]

ROBUST FIELD-LEVEL LIKELIHOOD-FREE INFERENCE WITH GALAXIES



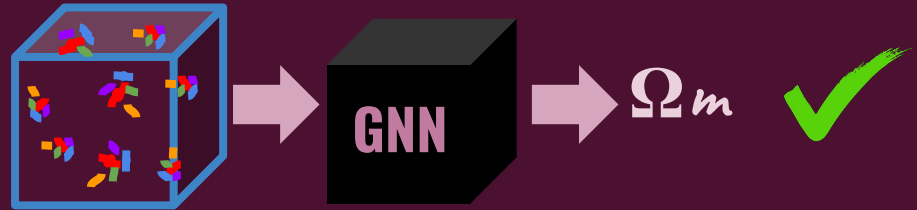
QUESTIONS

- 1 - What are the galaxy fundamental properties to infer Ω_m using a GNN?

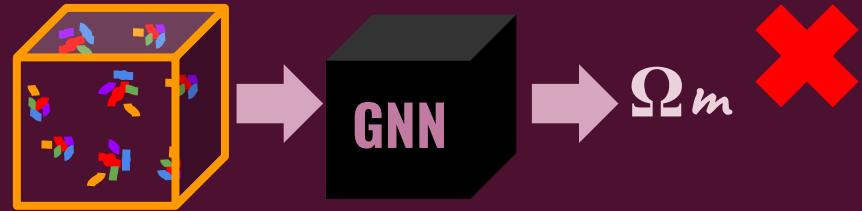


QUESTIONS

- 1 - What are the galaxy fundamental properties to infer Ω_m using a GNN?

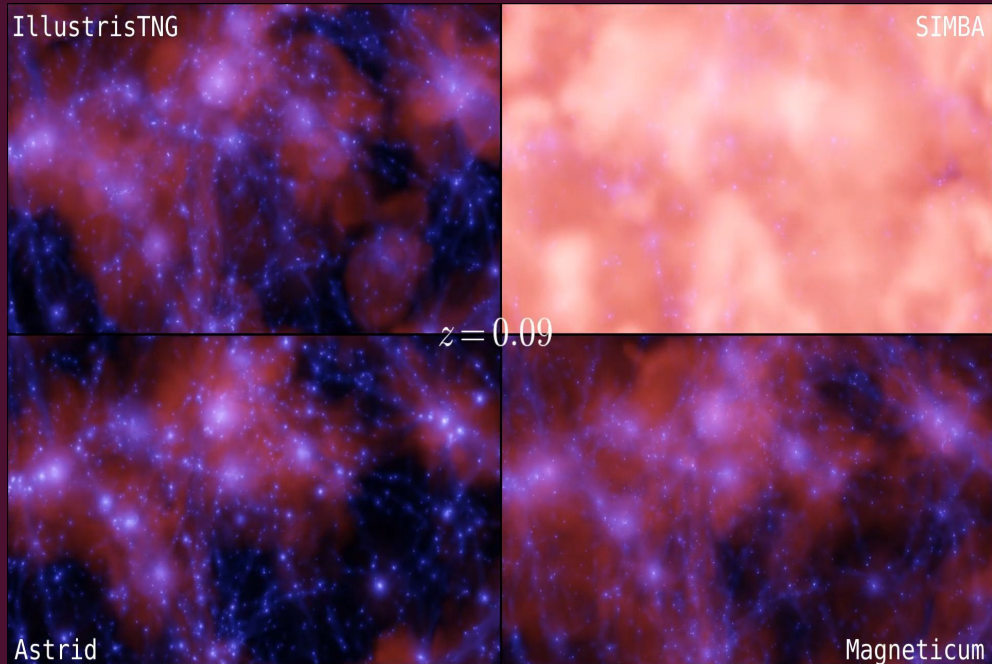


- 2 - Is it possible to get a robust model?

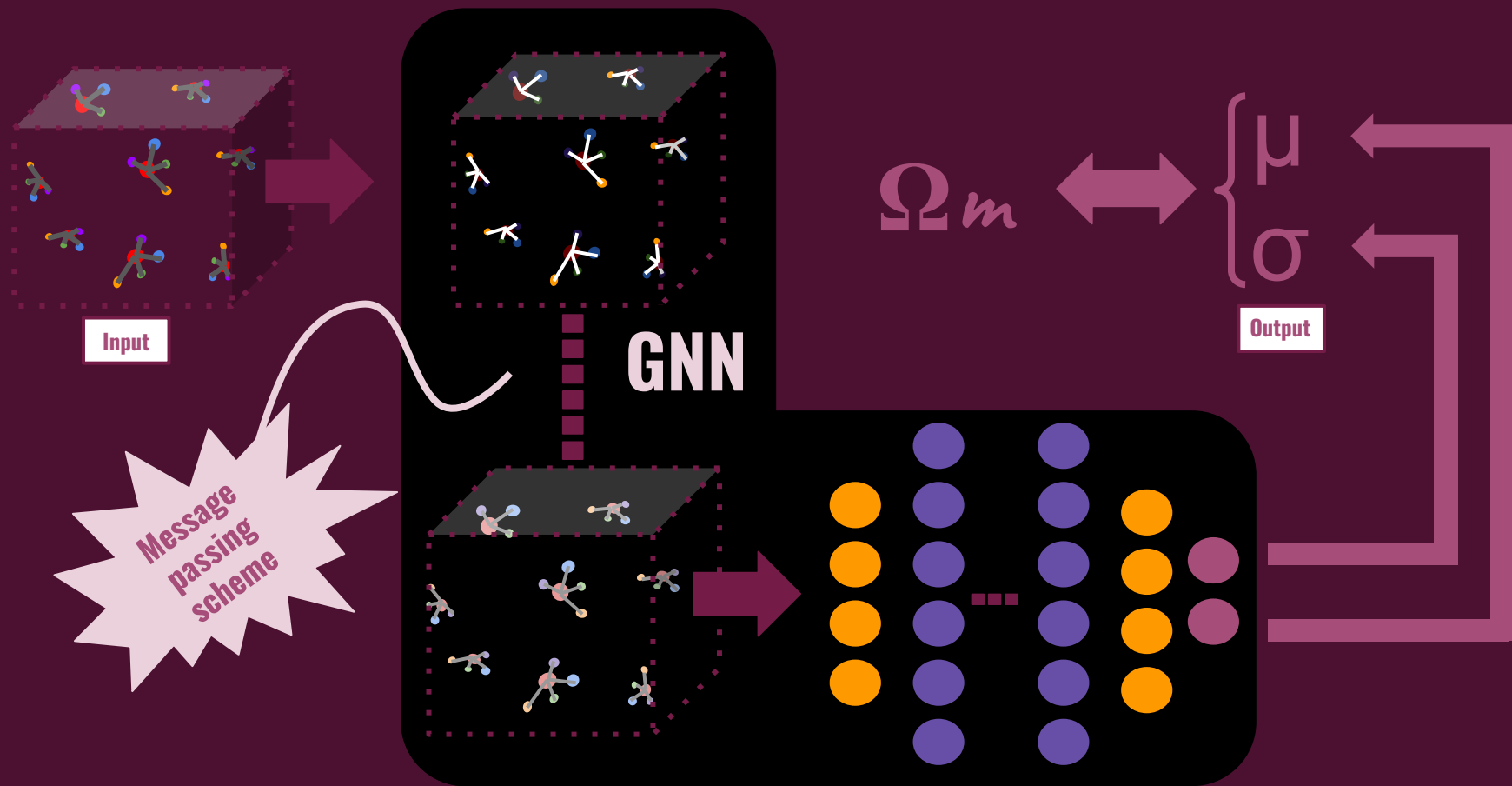


QUESTIONS

- 1 - What are the galaxy fundamental properties to infer Ω_m using a GNN?
- 2 - Is it possible to get a robust model?



CAMELS visualizations

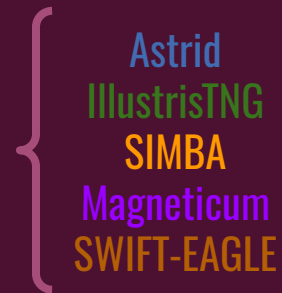


THE JOURNEY

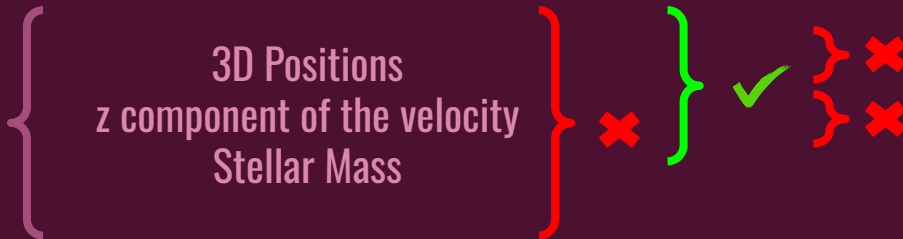
Training set



Testing sets

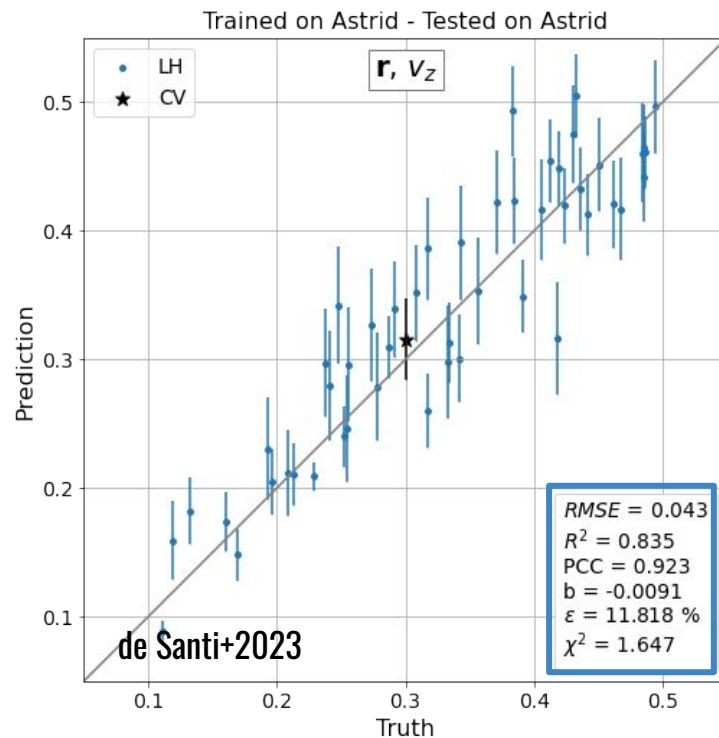


Galaxy Information

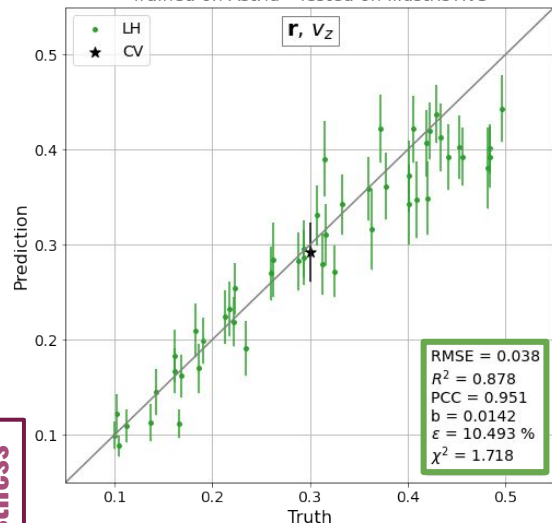


THE BEST MODEL

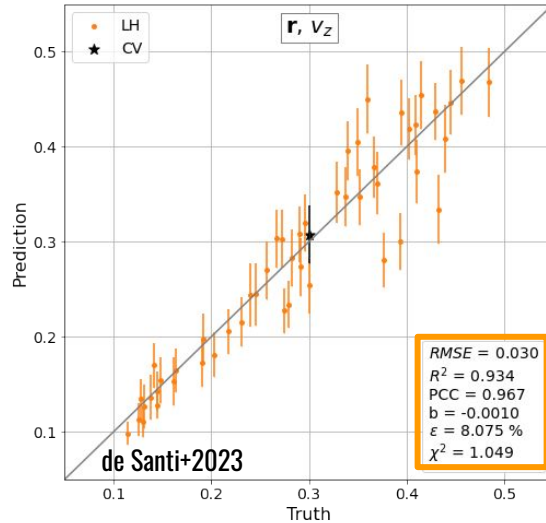
Sanity check



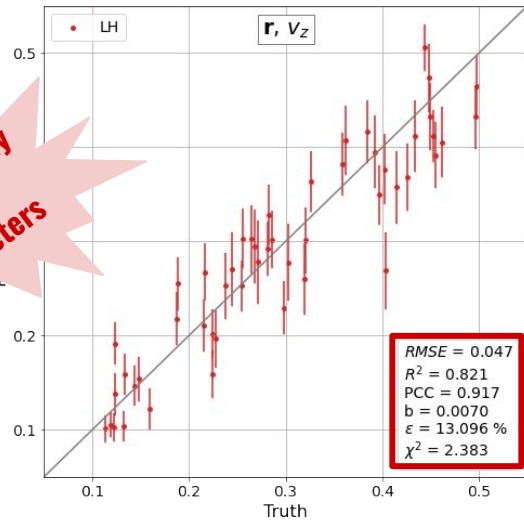
Trained on Astrid - Tested on IllustrisTNG



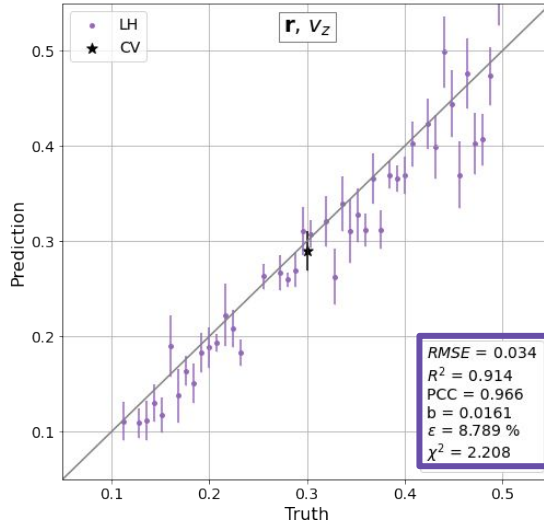
Trained on Astrid - Tested on SIMBA



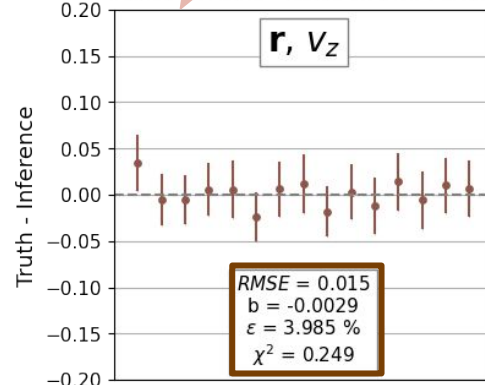
Trained on Astrid - Tested on SB28



Trained on Astrid - Tested on Magneticum



Trained on Astrid - Tested on SWIFT-EAGLE



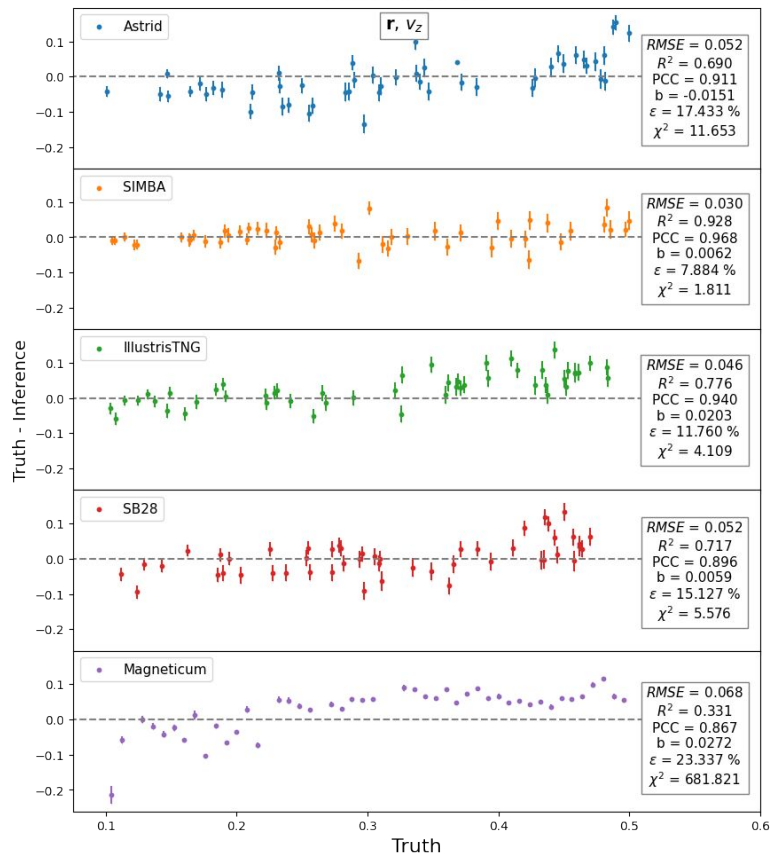
Using
halos/subhalos
found with a
different finder

Robustness

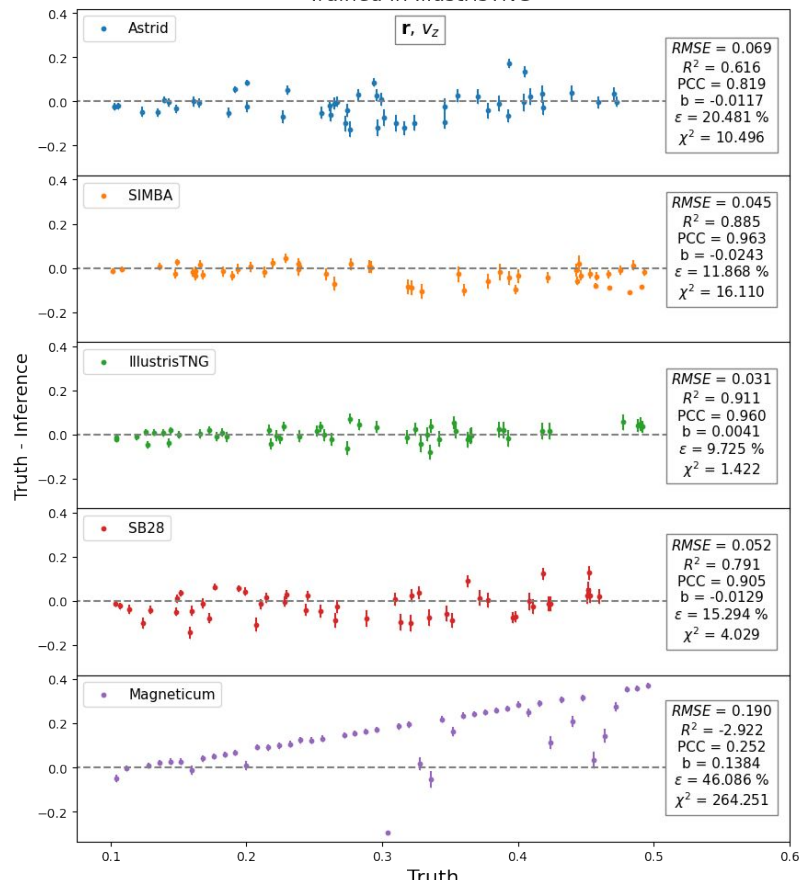
Varying many
other
parameters

Training the model using different simulations

Trained in SIMBA

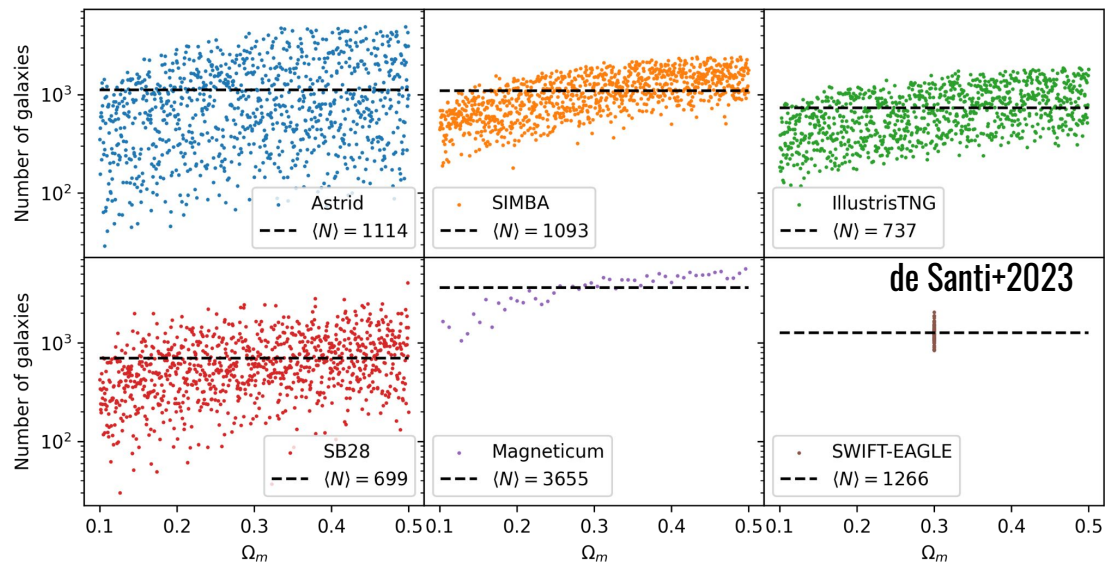


Trained in IllustrisTNG



Why the model trained on Astrid is so good?

- We have a **broader variation** in the **number of galaxies** in **Astrid** catalogs

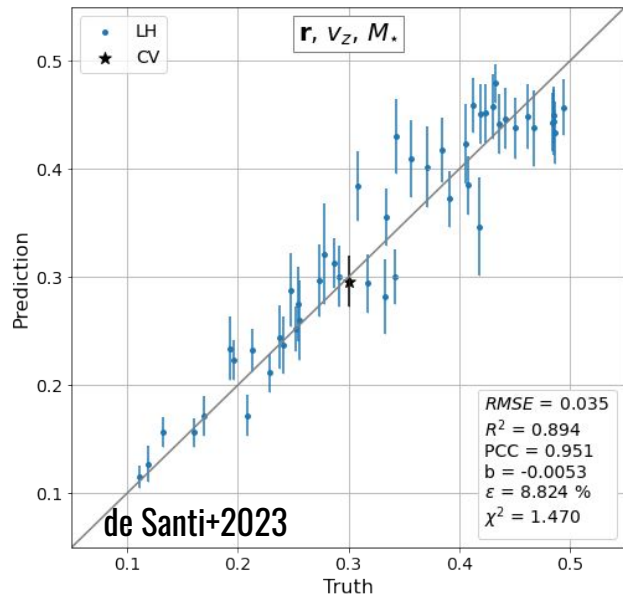


- Astrid** produces **larger variation in some galaxy properties** given the parameter variations in the LH set

Trying to improve the model

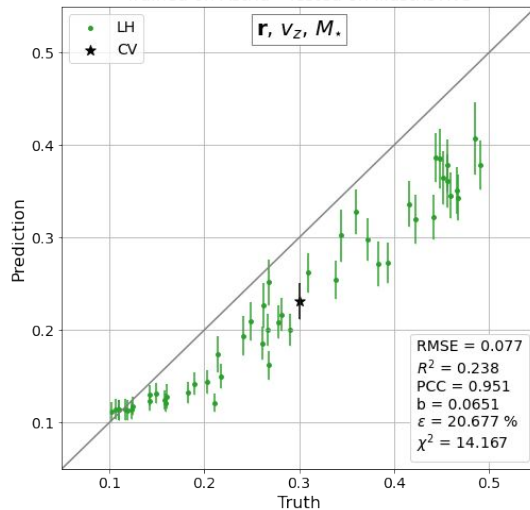
Sanity check

Trained on Astrid - Tested on Astrid

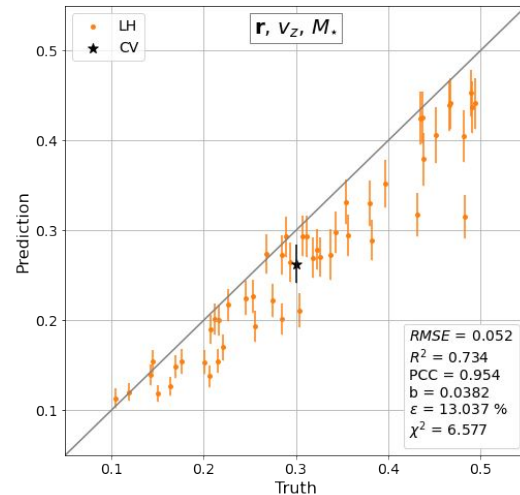


Robustness

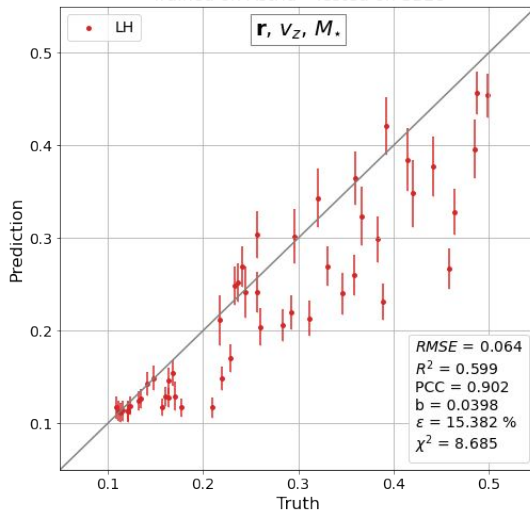
Trained on Astrid - Tested on IllustrisTNG



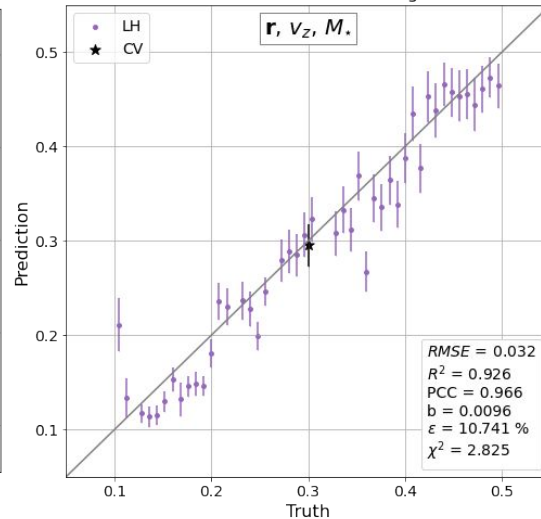
Trained on Astrid - Tested on SIMBA



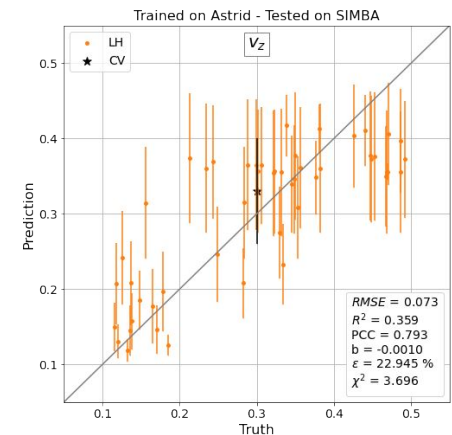
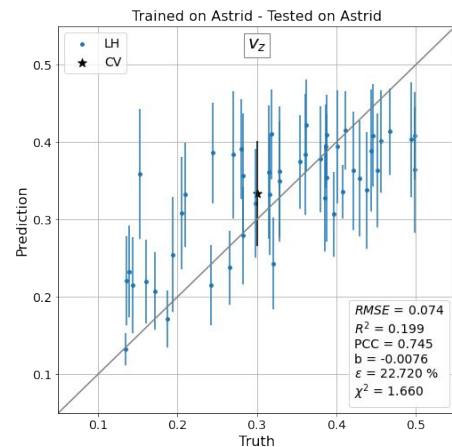
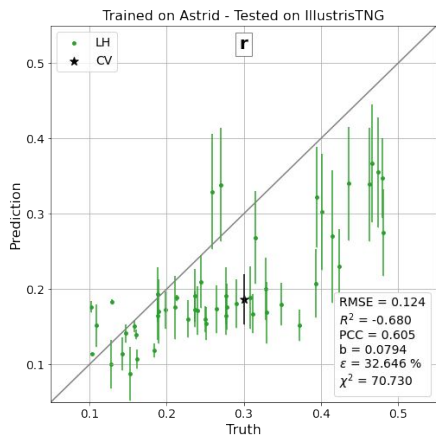
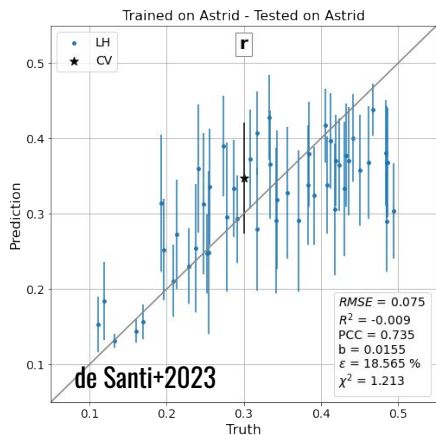
Trained on Astrid - Tested on SB28



Trained on Astrid - Tested on Magenticum



Where does the information come from?



Positions

Velocities

GNN without
node
information

Sanity check

Robustness

We need both!

Error bars
increased

Breaks down

Deep Set

CONCLUSIONS

1 - The most important galaxy properties for the GNN

constrain Ω_m :

galaxy positions and velocities (z component)

2 - We can constrain Ω_m for:

5 different hydrodynamical simulations

	ϵ	χ^2
Astrid	11.8 %	1.6
SIMBA	8.1 %	1.0
IllustrisTNG	10.5 %	1.7
SB28	13.0 %	2.4
Magneticum	8.8 %	2.2
SWIFT-EAGLE	4.0 %	0.3

Hands-on activity

STEP 1: go to github and download the repo using:

```
$ git clone https://github.com/natalidesanti/pytorch and GNNs
```

STEP 2: open the notebook in your preferred machine (or using Google Colab)

